Theo Farrell

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Durham University MSci Natural Sciences student specialising in mechanistic interpretability of neural networks. Won L2 and L3 Natural Sciences Outstanding Achievement Awards. Founded Durham's AI Safety Initiative and secured an $6,500 grant from Coefficient Giving in 2025 for independent research. Recent L4 project research work at Durham University resulted in first-authorship of a paper for a NeurIPS 2025 workshop.

EDUCATION

Durham University 2022 – 2026
MSci Natural Sciences (Computer Science & Philosophy) Average grade: 1st (77%, 4.0 GPA)

- **Dissertation:** "Relational Composition and Sparse Autoencoders", supervised by Noura Al Moubayed.
- **Key Modules:** NLP, Deep Learning, Computer Vision, Reinforcement Learning.
- **Annual End-of-Year Averages:** Y1: 76%, Y2: 76.5%, Y3: 77%.
- **Top Module Rankings:** Natural Computing (1st/51), Computer Vision (2nd/81), Moral Theory (1st/135), Political Philosophy (2nd/138).

A-Levels Results 2020 – 2022
4 A-Levels (A*A*A*A*) including Further Maths, in addition to EPQ (A*).

EXPERIENCE

University of Calgary 06/2024 – 09/2024
MITACS Global Research Intern (supervised by Aditya Nittala)

- Co-author on FabFoam (2025): used Arduino and Java to create and program soft electrotactile devices that provide haptic feedback.
- Designed a VR application using A-Frame (HTML and JavaScript) on the Meta Quest 3 for a paper at ACM UIST'24 (acknowledged, `https://doi.org/10.1145/3654777.3676448`).

Founder and Lead Organiser 09/2023 – Ongoing
Durham AI Safety Initiative

- Organised recruitment and increased our weekly attendance from 5 to 20 participants.
- Facilitated BlueDot discussion groups to upskill members in AI alignment and policy knowledge.
- Led technical upskilling workshops in deep learning and NLP based on ARENA.
- Secured grant funding for group organising.
- Pathfinder Fellow since Dec 2024 - received training and mentorship for AI Safety organising after a competitive application process.

PUBLICATIONS [Scholar]

- Farrell, Theo, Patrick Leask, and Noura Al Moubayed. "Order by Scale: Relative-Magnitude Relational Composition in Attention-Only Transformers." Socially Responsible and Trustworthy Foundation Models at NeurIPS 2025. Link: `https://openreview.net/forum?id=vWRVzNtk7W`
- Kempermann, Manon, et al. "Challenges of Evaluating LLM Safety for User Welfare." Second Conference of the International Association for Safe and Ethical Artificial Intelligence (IASEAI'26), 2026. Link (preprint): `https://arxiv.org/abs/2512.10687`

Reviewing Activities: NeurIPS 2025 Mechanistic Interpretability Workshop, NeurIPS 2025 ResponsibleFM Workshop.

GRANTS AND AWARDS

- **$4,500 from Pathfinder** (10/2025): 1 year of group expenses for Durham AI Safety Initiative.
- **$6,500 from Coefficient Giving (fmr. Open Philanthropy)** (07/2025): 4 months of independent mechanistic interpretability research.

- **$8,000 from Coefficient Giving (fmr. Open Philanthropy)** (03/2025): 1 year of group expenses for Durham AI Safety Initiative.
- **L3 Natural Sciences Outstanding Achievement** (07/2025): Averaged over 75% in 3rd year.
- **L2 Natural Sciences Outstanding Achievement** (07/2024): Averaged over 75% in 2nd year.
- **Gold Duke of Edinburgh Award** (2022).

CERTIFICATIONS

- **AGI Strategy** (BlueDot Impact, 2026): 30-hour intensive covering technical AI trends and future capabilities, threat modelling via kill chain analysis, and defence-in-depth frameworks for designing protective interventions. [Certificate]

TECHNICAL SKILLS

Programming (Fluent)	Python, C#, Javascript, HTML.
Programming (Familiar)	C, C++, Java, Haskell, Fortran, Ruby.
Tools	PyTorch, TensorFlow, scikit-learn, Hugging Face, NumPy, Weights & Biases, TransformerLens, OpenCV, Seaborn, matplotlib, pandas, Arduino, Git, uv, LaTeX, AR/VR, Unity.